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Experiment No. 2 : Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

Aim

To analyze and implement contrast enhancement techniques using histogram analysis and histogram equalization in Python and OpenCV for improving image quality and enhancing visual information in low-contrast images.

Purpose

Image contrast plays a significant role in the performance of computer vision systems. Poor illumination, sensor limitations, or environmental conditions often result in low-contrast images, making feature extraction and object recognition difficult. Histogram-based enhancement techniques improve the distribution of pixel intensities, thereby enhancing image visibility and facilitating subsequent image analysis tasks. In this experiment, students will implement contrast stretching, histogram analysis, and histogram equalization using Python and OpenCV. The effectiveness of these enhancement techniques will be evaluated on real-world images such as medical images, satellite images, or low-light photographs.

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

# Step 1 & 2: Load and convert to Grayscale
image_path = "/content/Screenshot 2026-06-21 222959.png" # Replace with your image file path
img_bgr = cv2.imread(image_path)

if img_bgr is None:
    raise FileNotFoundError("Image not found. Please provide a valid path.")

img_gray = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)

# Step 4: Contrast Stretching (Min-Max Normalization to [0, 255])
min_val = np.min(img_gray)
max_val = np.max(img_gray)
if max_val > min_val:
    img_stretched = ((img_gray - min_val) / (max_val - min_val) * 255.0).astype(np.uint8)
elif min_val == max_val:
    # Handle cases where all pixels have the same value to avoid division by zero
    img_stretched = img_gray.copy()
else:
    img_stretched = img_gray.copy()

# Step 5: Global Histogram Equalization (HE)
img_he = cv2.equalizeHist(img_gray)

# Step 6: Contrast Limited Adaptive Histogram Equalization (CLAHE)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
img_clahe = clahe.apply(img_gray)

# Step 7: Calculate Histograms
hist_orig = cv2.calcHist([img_gray], [0], None, [256], [0, 256])
hist_stretched = cv2.calcHist([img_stretched], [0], None, [256], [0, 256])
hist_he = cv2.calcHist([img_he], [0], None, [256], [0, 256])
hist_clahe = cv2.calcHist([img_clahe], [0], None, [256], [0, 256])

# Visualization: Images and Histograms Comparison
fig, axes = plt.subplots(4, 2, figsize=(12, 16))

# 1. Original
axes[0, 0].imshow(img_gray, cmap="gray")
axes[0, 0].set_title("1. Original Grayscale Image")
axes[0, 1].plot(hist_orig, color="black")
axes[0, 1].set_title("Original Histogram")
axes[0, 1].set_xlim([0, 256])

# 2. Contrast Stretching
```

```
axes[1, 0].imshow(img_stretched, cmap="gray")
axes[1, 0].set_title("2. Contrast Stretched Image")
axes[1, 1].plot(hist_stretched, color="blue")
axes[1, 1].set_title("Contrast Stretched Histogram")
axes[1, 1].set_xlim([0, 256])

# 3. Global Histogram Equalization
axes[2, 0].imshow(img_he, cmap="gray")
axes[2, 0].set_title("3. Global Histogram Equalized Image")
axes[2, 1].plot(hist_he, color="red")
axes[2, 1].set_title("Equalized Histogram (Flattened)")
axes[2, 1].set_xlim([0, 256])

# 4. CLAHE
axes[3, 0].imshow(img_clahe, cmap="gray")
axes[3, 0].set_title("4. CLAHE Enhanced Image")
axes[3, 1].plot(hist_clahe, color="green")
axes[3, 1].set_title("CLAHE Histogram")
axes[3, 1].set_xlim([0, 256])

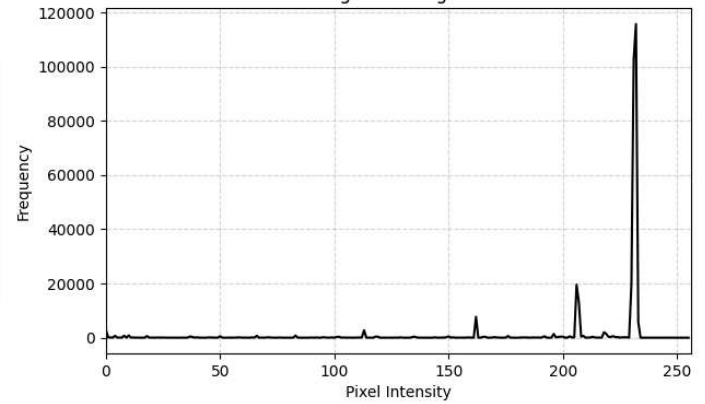
# Formatting
for row in axes:
    row[0].axis("off")
    row[1].grid(True, linestyle="--", alpha=0.5)
    row[1].set_xlabel("Pixel Intensity")
    row[1].set_ylabel("Frequency")

plt.tight_layout()
plt.show()
```

1. Original Grayscale Image



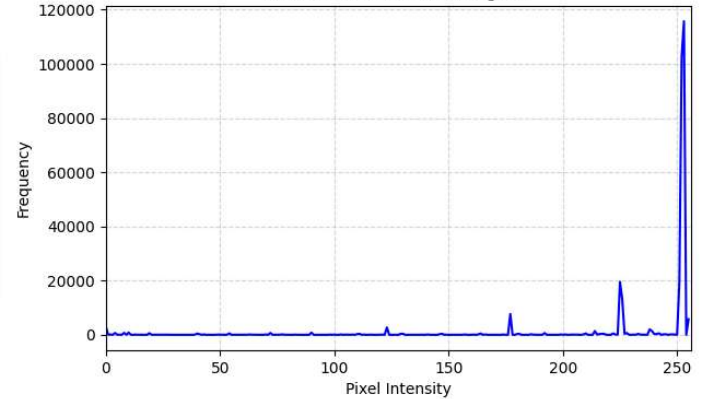
Original Histogram



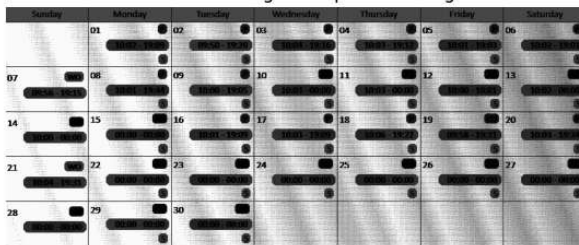
2. Contrast Stretched Image



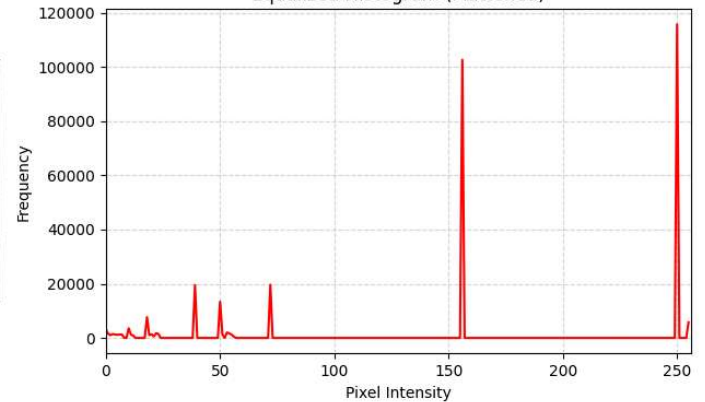
Contrast Stretched Histogram



3. Global Histogram Equalized Image



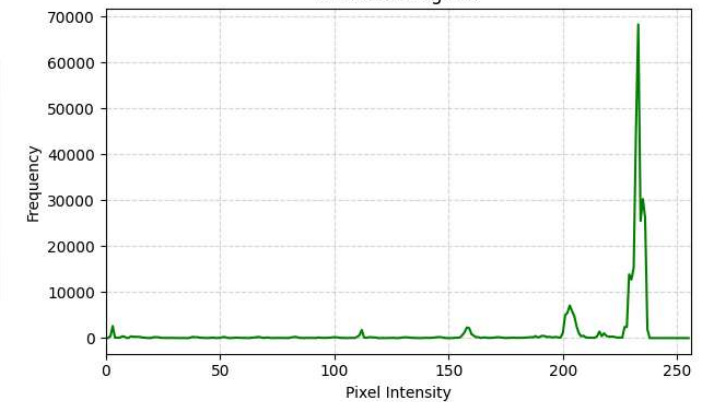
Equalized Histogram (Flattened)



4. CLAHE Enhanced Image



CLAHE Histogram



Answers to Experiment Questions

1. What is image contrast, and why is it important in image processing?

Image Contrast is the difference in luminance or color intensity that makes objects and structures distinguishable from each other and from the background. Mathematically, it reflects the dynamic range between maximum and minimum intensity values.

Importance: High contrast clarifies edge boundaries, textures, and subtle details, which is critical for accurate image binarization, edge detection, segmentation, and machine learning feature extraction.

2. Explain the concept of an image histogram. What information does it provide?

An image histogram is a graphical representation of the tonal distribution in a digital image. It plots the frequency (number of pixels) on the y-axis against intensity bins (0 to 255 for 8-bit images) on the x-axis.

Information Provided:

- **Exposure & Dynamic Range:** Indicates whether an image is under-exposed (data skewed left toward 0), over-exposed (skewed right toward 255), or low-contrast (clustered in a narrow central band).
- **Bimodality:** Shows whether distinct foreground and background peaks exist, assisting in choosing optimal thresholding values (e.g., Otsu's method).

3. Differentiate between Histogram Stretching and Histogram Equalization.

- **Histogram Stretching (Min-Max Normalization):** A linear transformation that scales existing pixel values to fill the full dynamic range $[0, 255]$. It preserves relative intensity gaps and histogram shape without altering monotonic relationships between bins.
- **Histogram Equalization (HE):** A non-linear transformation based on the Cumulative Distribution Function (CDF). It flattens the histogram to achieve an approximately uniform probability density across all intensity levels, redistributing clustered values across the entire spectrum.

4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

CLAHE is an adaptive, local enhancement method. It divides an image into small contextual blocks or tiles (e.g., 8×8) and applies histogram equalization locally to each tile.

Difference from Global HE:

- **Local Processing:** Standard HE computes a single global CDF across the entire image, whereas CLAHE calculates localized CDFs to preserve local details.
- **Contrast Limiting:** CLAHE clips the histogram at a predefined limit (clipLimit) before computing the CDF, redistributing excess pixels uniformly across all bins to prevent local noise and artifact amplification. Bilinear interpolation is then applied across tile borders to remove artificial boundary seams.

5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

- **Illumination Invariance:** Standardizes global lighting conditions, reducing variations caused by lighting angles, sensor sensitivity, or shadows.
- **Sharper Boundary Gradients:** Enhances gradient magnitudes at object edges, leading to higher accuracy during edge detection (Canny/Sobel), thresholding, and contour extraction.

6. Mention three real-world applications where histogram equalization is widely used.

- **Medical Imaging:** Enhancing low-contrast tissue distinctions in X-rays, mammograms, CT scans, and MRI data.
- **Satellite & Remote Sensing:** Enhancing terrain details, vegetation boundaries, and water bodies in aerial multispectral imagery.
- **Underwater & Night Vision Photography:** Mitigating severe light attenuation and backscatter in underwater images and low-light surveillance feeds.

7. What are the limitations of global histogram equalization?

- **Noise Amplification:** Over-amplifies background noise and sensor grain in homogeneous or near-flat regions.
- **Loss of Local Detail:** Strong global intensity peaks can wash out fine local details in darker or brighter sub-regions.
- **Unnatural Artifacts:** Can introduce harsh over-saturation, halo artifacts, and posterization effects across natural photographic scenes.